



# Covariate Adjustment

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CML Week 2

2026-09-08

# Schedule

1. Design-based causal inference & Monte Carlo simulation
2. **Covariate adjustment**
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference

# Readings for Today

## Foundational

- › ([Freedman 2008](#)): On regression adjustments to experimental data

## Common

- › ([Bai, Shaikh, and Tabord-Meehan 2024](#)): A primer on the analysis of randomized experiments

## Presenter

- › ([Lin 2013](#)): Agnostic notes on regression adjustments
- › ([Aronow and Middleton 2013](#)): A class of unbiased estimators of the ATE
- › ([Belloni, Chernozhukov, and Hansen 2014](#)): Inference after selection among high-dimensional controls
- › ([Wager et al. 2016](#)): High-dimensional adjustments in randomized experiments



# The Central Question

| *Should we adjust for covariates in randomized experiments?*

- › **Classical View:** Adjustment improves precision
- › **Skeptical View:** Model misspecification can introduce bias
- › **Modern View:** Principled approaches can achieve both goals

# (Freedman 2008): The Critique

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# The Freedman Position

- › Randomization ensures unbiased estimates **without adjustment**
- › Regression adjustment assumes a **model for the outcome**
- › Regression adjustment is **biased** — but only at order  $1/n$
- › The lasting damage is to **variance** and to **standard errors**
- › “The cure can be worse than the disease”

Does that mean we should just stick with basic differences-in-means?

# Key Arguments

“Regression adjustments are often made to experimental data. Since randomization does not justify the models, almost anything can happen.”

- › Models are **not justified** by the experimental design
- › Even **simple linear models** can fail badly
- › **Interaction effects** are typically ignored
- › **Heteroskedasticity** complicates inference

# Freedman's Setup

He does not posit a wrong model. He rewrites  $Y_i$  as a regression — and this is an **identity**, not an assumption. Write  $\mu_a = \frac{1}{N} \sum_i Y_i(a)$  and  $\pi = N_1/N$ . Then for every unit,

$$Y_i = \underbrace{\pi\mu_1 + (1 - \pi)\mu_0}_a + \tau(A_i - \pi) + \delta_i, \quad \delta_i = \alpha_i + (\tau_i - \tau)(A_i - \pi)$$

- ›  $\alpha_i = \pi(Y_i(1) - \mu_1) + (1 - \pi)(Y_i(0) - \mu_0)$ : unit  $i$ 's **level**
  - » averaging over the randomization distribution
- ›  $\tau_i - \tau$ : unit  $i$ 's **effect heterogeneity**
  - » the quantity behind  $s_\tau^2$  in Week 1

Both are centred **by construction** —  $\mu_a$  is the mean of  $Y_i(a)$ ,  $\tau$  the mean of  $\tau_i$ :

$$\sum_i \alpha_i = \sum_i (\tau_i - \tau) = 0$$

*Freedman writes  $T_i, C_i, p, Z$  where we write  $Y_i(1), Y_i(0), \pi, X$ .*



# Why That Identity Is Damning

$$\delta_i = \alpha_i + (\tau_i - \tau)(A_i - \pi)$$

“Equation (4) is nothing like a standard regression model. The randomness in  $\delta_i$  is due entirely to randomness in  $[A_i]$ , so the error term is strongly dependent on the explanatory variable.”

$\alpha_i$  and  $\tau_i$  are **fixed numbers**. The only randomness in  $\delta_i$  is  $A_i$  — which is the regressor. So conditioning on the design pins the error down exactly:

$$E[\delta_i | \mathbf{A}] = \delta_i \neq 0$$

That is **strict exogeneity** (a key condition of the Gauss-Markov theorem) failing. This means we do not have an unbiased estimator



# How Severe Is This?

- › The  $\delta_i$  are **not i.i.d.** and not individually mean-zero — though  $\frac{1}{N} \sum_i \alpha_i = 0$ , so they do average out across units
- › The whole channel is effect heterogeneity: if  $\tau_i \equiv \tau$ , the second term vanishes and the problem goes away
- › The damage is **real but shrinking**: we will see next that the bias is  $O(1/N)$
- › In his simulations, bias is “quite noticeable” at  $N = 250$  and “**negligible**” by  $N = 1000$ 
  - » “Adjustment may help or hurt”
  - » nominal variances are off “by factors that are quite large”

**Small is not zero.** Difference-in-means is **exactly unbiased at every  $N$**  (last week).

- › There is no theoretical argument that adjustment *always* beats difference-in-means
  - » you are trading a finite-sample *guarantee* for an asymptotic *approximation*



# Proof Sketch: Where the $O(1/N)$ Bias Comes From

**Setup.** Regress  $Y_i$  on  $1, A_i - \pi$ , centred  $X_i$ . By FWL,  $\tilde{A}_i$  the residual of  $A_i - \pi$  on  $X_i$ :

$$\hat{\tau} - \tau = \frac{\sum_i \tilde{A}_i \delta_i}{\sum_i \tilde{A}_i^2}, \quad \tilde{A}_i = (A_i - \pi) - \hat{c} X_i, \quad \hat{c} = \frac{\sum_i X_i (A_i - \pi)}{\sum_i X_i^2}$$

**Without  $X$ :** both  $\sum_i (A_i - \pi) = 0$  and  $\sum_i (A_i - \pi)^2 = N\pi(1 - \pi)$  are **deterministic**: constant denominator, mean-zero numerator, **exactly unbiased**.

**With  $X$ :** substituting  $\delta_i = \alpha_i + (\tau_i - \tau)(A_i - \pi)$  splits the numerator in three. The first two have mean zero, by randomisation and by  $E[\hat{c}] = 0$ :

$$E\left[\sum_i \tilde{A}_i \delta_i\right] = E\left[\underbrace{\sum_i (A_i - \pi) \delta_i}_0\right] - E[\hat{c}] \underbrace{\sum_i X_i \alpha_i}_0 - E\left[\hat{c} \sum_i X_i (\tau_i - \tau)(A_i - \pi)\right]$$



# Proof Sketch: The Surviving Term

Both factors are linear in  $A$ , so this is a double sum over the assignment covariances:  $\pi(1 - \pi)$  on the diagonal,  $-\pi(1 - \pi)/(N - 1)$  off it:

$$\begin{aligned} \mathbb{E}\left[\hat{c} \sum_i X_i(\tau_i - \tau)(A_i - \pi)\right] &= \frac{1}{\sum_i X_i^2} \sum_j \sum_i X_j X_i(\tau_i - \tau) \mathbb{E}[(A_j - \pi)(A_i - \pi)] \\ &= \frac{\pi(1 - \pi)}{\sum_i X_i^2} \left[ \sum_i X_i^2(\tau_i - \tau) - \frac{1}{N - 1} \sum_{i \neq j} X_i X_j(\tau_i - \tau) \right] \\ &= \frac{N}{N - 1} \pi(1 - \pi) \frac{\sum_i X_i^2(\tau_i - \tau)}{\sum_i X_i^2} \end{aligned}$$

Centring collapses the cross terms:  $\sum_{j \neq i} X_j = -X_i$ , so the off-diagonal sum is the diagonal one. Dividing by  $\sum_i \tilde{A}_i^2 \approx N\pi(1 - \pi)$ :

$$\text{Bias}(\hat{\tau}) \approx -\frac{1}{N - 1} \cdot \frac{\sum_i X_i^2(\tau_i - \tau)}{\sum_i X_i^2}$$



# What about variance?

Standardise a covariate so  $\bar{X} = 0$  and  $\frac{1}{N} \sum_i X_i^2 = 1$ , then ask **two separate questions** about it:

$$c_{\text{lvl}} = \frac{1}{N} \sum_i \alpha_i X_i, \quad c_{\text{het}} = \frac{1}{N} \sum_i (\tau_i - \tau) X_i$$

Does  $X$  predict the **level**, and does it predict the **effect**?

Comparing Freedman's Theorems 3 and 4, the reduction in asymptotic variance from adjusting is

$$\frac{\Delta}{N\pi(1-\pi)}, \quad \Delta = c_{\text{lvl}} [c_{\text{lvl}} + 2(1-2\pi) c_{\text{het}}]$$

- ›  $\pi = 1/2$ :  $\Delta = c_{\text{lvl}}^2 \geq 0$  — adjustment is **neutral or helps**, always
- ›  $\pi \neq 1/2$ :  $\Delta$  can be **negative** — adjustment *raises* the asymptotic variance

**It takes both:** unequal arms **and** a covariate that predicts heterogeneity. Under the sharp null  $\tau_i \equiv 0$  we have  $c_{\text{het}} = 0$ , and adjustment can never hurt.



# The Conservative Conclusion

## Freedman's Recommendation:

- › Use **difference-in-means** (no covariate adjustment)
- › Randomization **guarantees unbiasedness**
- › Avoid model-dependent estimators in experiments

# Discussion: Freedman's Critique

1. The bias Freedman finds vanishes at rate  $1/n$ . Does that make this a **small-sample warning**, or something more fundamental? Which reading do you buy?
2. Freedman was writing about the experiments of his era. Would he write the same paper about an **A/B test with a million users**? What actually changes?
3. You referee a paper reporting only the *adjusted* estimate. Another reports only the *unadjusted* one. **Which do you trust less, and why?**



## (Lin 2013): Agnostic Responses

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# Responding to Freedman

## Lin's Goal:

- › Find **robust approaches** to covariate adjustment
- › **Agnostic methods** that don't rely on strong modeling assumptions
- › Preserve the benefits of adjustment while minimizing risks

# The Agnostic Philosophy

- › Start with **difference-in-means** as baseline
- › Add covariates in a “**model-free**” manner
- › Focus on **asymptotic robustness**

# Lin's Insight

Consider the **saturated model**:

$$Y_i = \alpha + \tau A_i + X_i' \beta + A_i X_i' \gamma + \varepsilon_i$$

## Key Properties:

- › **Always includes interactions** between treatment and covariates
- › Covariates are **centered**
- › **Consistent for the ATE** regardless of  $\beta$  and  $\gamma$
- › **More efficient** than difference-in-means when covariates predict outcomes



# Why Saturation Helps

**Theorem (Lin):** the interacted estimator is **consistent** for the ATE, and its asymptotic variance is **never larger** than the difference-in-means — whatever the true relationship between  $Y$  and  $X$ .

Why: Lin's estimator is the difference-in-means applied to **residualised** potential outcomes,  $e_i(a) = Y_i(a) - \mu_a - X_i b_a$ , with  $b_a$  the arm- $a$  slope. So the Week 1 variance applies unchanged — to  $e$  rather than  $Y$ :

$$\frac{1}{N} \left[ \frac{\sigma_1^2}{\pi} + \frac{\sigma_0^2}{1-\pi} - \sigma_\tau^2 \right] \rightarrow \frac{1}{N} \left[ \frac{\sigma_{e1}^2}{\pi} + \frac{\sigma_{e0}^2}{1-\pi} - \sigma_{e\tau}^2 \right]$$



# Proof Sketch: The Gain Is a Square

With  $X$  centred and standardised,  $\sigma_{\epsilon\alpha}^2 = \sigma_\alpha^2 - b_\alpha^2$  and  $\sigma_{\epsilon\tau}^2 = \sigma_\tau^2 - (b_1 - b_0)^2$ :

$$\begin{aligned} V_{\text{unadj}} &= \frac{1}{N} \left[ \frac{\sigma_1^2}{\pi} + \frac{\sigma_0^2}{1-\pi} - \sigma_\tau^2 \right] \\ V_{\text{Lin}} &= \frac{1}{N} \left[ \frac{\sigma_1^2 - b_1^2}{\pi} + \frac{\sigma_0^2 - b_0^2}{1-\pi} - \sigma_\tau^2 + (b_1 - b_0)^2 \right] \\ V_{\text{unadj}} - V_{\text{Lin}} &= \frac{1}{N} \left[ \frac{b_1^2}{\pi} + \frac{b_0^2}{1-\pi} - (b_1 - b_0)^2 \right] \\ &= \frac{1}{N} \left[ \frac{1-\pi}{\pi} b_1^2 + 2b_1b_0 + \frac{\pi}{1-\pi} b_0^2 \right] \\ &= \frac{[(1-\pi)b_1 + \pi b_0]^2}{N\pi(1-\pi)} \geq 0 \end{aligned}$$



# Lin Completes Freedman's Square

Since  $c_{lvl} = \pi b_1 + (1 - \pi)b_0$  and  $c_{het} = b_1 - b_0$ , we have  $(1 - \pi)b_1 + \pi b_0 = c_{lvl} + (1 - 2\pi)c_{het}$ . Over a common  $N\pi(1 - \pi)$ , the two gains are

$$\underbrace{c_{lvl} [c_{lvl} + 2(1 - 2\pi)c_{het}]}_{\text{un-interacted (Freedman)}} \quad \text{vs.} \quad \underbrace{[c_{lvl} + (1 - 2\pi)c_{het}]^2}_{\text{interacted (Lin)}}$$

differing by exactly  $(1 - 2\pi)^2 c_{het}^2 \geq 0$ . Freedman's expression is an **incomplete square**; Lin supplies the missing term.

- › Vanishes at  $\pi = 1/2$  — with equal arms, interacting buys nothing asymptotically
- › Vanishes when  $c_{het} = 0$  — no heterogeneity for  $X$  to pick up, which is also when Freedman's  $\Delta$  could not go negative



# Practical Recommendations

1. **Always use saturated models** (include interactions)
2. **Use heteroskedasticity-robust standard errors**
3. **Pre-specify covariate adjustment plans**
4. **Report both adjusted and unadjusted estimates**
5. **Check sensitivity** to specification choices

# HC2 Standard Errors

Lin shows that the **appropriate standard errors** are:

$$\text{Var}(\hat{\tau}) \approx \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0}$$

where  $S_1^2$  and  $S_0^2$  are sample variances of residuals in each group.

This is the **HC2** (heteroskedasticity-consistent) estimator we talked about last week!



# Discussion: Lin's Agnostic Approach

1. “Agnostic” about the **model**, but not about the **design**. What is Lin still assuming — and where would you expect it to break first?
2. Lin says *always interact*. Bai et al. note that at  $\pi = 1/2$  the un-interacted regression has the **same asymptotic variance** with fewer parameters. What would you tell a practitioner?
3. The guarantee is asymptotic. Field experiment,  $n = 60$ , 12 covariates, interactions included. **Do you believe it?** What would convince you either way?



# (Aronow and Middleton 2013): Unbiased Estimators

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# A Unifying Framework: Horvitz–Thompson

**Main result:** if you *know* the assignment probabilities, unbiasedness is available for any design. Covariate adjustment comes free on top.

Start from Horvitz–Thompson. With  $\pi_{1i} = \Pr[A_i = 1]$  and  $\pi_{0i} = \Pr[A_i = 0]$  known by design:

$$\hat{\tau}_{\text{HT}} = \frac{1}{N} \sum_i \left[ \frac{A_i Y_i}{\pi_{1i}} - \frac{(1 - A_i) Y_i}{\pi_{0i}} \right]$$

Unbiased because  $E[A_i] = \pi_{1i}$  — exactly, at every  $N$ , with no model and no asymptotics.

- › Holds when propensities **differ across units**, and under **clustered** assignment
- › Both are settings where difference-in-means is **biased**



# Adjustment for Free

**The move:** subtract a predetermined  $f(X_i, \theta_i)$  from every outcome before weighting — a *generalized difference* estimator:

$$\hat{\tau}_G = \frac{1}{N} \sum_i \left[ \frac{A_i(Y_i - f(X_i, \theta_i))}{\pi_{1i}} - \frac{(1 - A_i)(Y_i - f(X_i, \theta_i))}{\pi_{0i}} \right]$$

$f$  can be **anything** — linear, logistic, a forest. In their words, the choice of  $f$

“may be relevant for efficiency, [but] it has no bearing on the unbiasedness of the estimator, so long as the choice is determined prior to examining the data.”

If  $f$  predicts  $Y$  well, the weighted terms co-vary across randomisations and the variance drops. If it predicts badly, you have lost nothing.



# Proof Sketch: Why $f$ Cancels

Take one unit, one arm. With  $f_i$  **fixed**,  $E[A_i] = \pi_{1i}$  gives

$$E \left[ \frac{A_i(Y_i - f_i)}{\pi_{1i}} \right] = \frac{\pi_{1i}(y_i(1) - f_i)}{\pi_{1i}} = y_i(1) - f_i$$
$$E \left[ \frac{(1 - A_i)(Y_i - f_i)}{\pi_{0i}} \right] = \frac{\pi_{0i}(y_i(0) - f_i)}{\pi_{0i}} = y_i(0) - f_i$$

Subtracting, the  $f_i$  cancels — whatever it was:

$$E[\hat{\tau}_G] = \frac{1}{N} \sum_i (y_i(1) - y_i(0)) = \tau$$

# When It Breaks

Fit  $\theta_i$  on the data and  $f(X_i, \theta_i)$  depends on the realised assignment, so  $E[A_i f_i] = \pi_{1i} E[f_i] + \text{Cov}(A_i, f_i)$ . Carrying that term through:

$$\text{Bias}(\hat{\tau}_G) = -\frac{1}{N} \sum_i \text{Cov}(A_i, f(X_i, \theta_i)) \left( \frac{1}{\pi_{1i}} + \frac{1}{\pi_{0i}} \right)$$

**Theorem (A&M):**  $\hat{\tau}_G$  is unbiased if  $\text{Cov}(A_i, f(X_i, \theta_i)) = 0$  for every  $i$ .

“If  $\theta_i$  is derived from the data, unbiasedness is not guaranteed, because the value of  $f(X_i, \theta_i)$  can depend on the particular randomization.”

The simplest way to satisfy it: take  $\theta_i$  from an **auxiliary or prior source**. Within linear  $f$  the ideal  $\theta_i$  is just the OLS fit of  $Y$  on  $X$  — so fit that OLS somewhere  $A_i$  cannot reach.



# Choosing $f$

| $f(X_i, \theta_i)$                       | What you get                                  |
|------------------------------------------|-----------------------------------------------|
| 0                                        | Plain Horvitz–Thompson                        |
| a constant $c$                           | HT recentred — already a variance improvement |
| $\theta_i' X_i$                          | Linear covariate adjustment                   |
| $1 - 1 / (1 + \exp(\theta_i' X_i))$      | Logistic, for binary $Y$                      |
| any fit from a pilot or prior experiment | ML adjustment, still <i>exactly</i> unbiased  |

**Key:** the choice of  $f$  is about **variance** alone. Unbiasedness never requires  $f$  to be *right* — only **fixed**.



# The Foundation for Advanced Methods

**Key Insight:** “Fixed, or from an independent source” is a condition you can *engineer*, not just hope for.

**In practice:**

- › Fit  $f$  on a **pilot study**, a prior experiment, or auxiliary survey data
- › **Pre-specify** the functional form before the outcomes are collected
- › **Sample splitting:** fit  $f$  on one fold, estimate the effect on another
- › **Cross-fitting:** rotate the folds so every unit contributes to both jobs

The last two are the modern move: they *manufacture* independence between the adjustment and the outcomes it is applied to, which is what lets arbitrarily flexible ML sit inside an unbiased estimator.



# Discussion: Data Independence

1. *Never let the adjustment see the outcomes* is a clean rule. **How often could you actually follow it** in work you have done or read?
2. Sample splitting buys back data independence and spends half your data to do it. **Why or why not would that be a good tradeoff?**
3. Pre-registration **socially** enforces discipline in choosing  $f$ . Is that a good enough substitute?

# (Bloniarz et al. 2016): Adjustment When $p$ Is Large

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# Where Lin's Estimator Runs Out of Room

**The hidden assumption:** the Freedman/Lin debate takes the covariate count  $p$  as small and fixed.

**Modern experiments do not:**

- › A/B tests carrying the whole behavioural log for each user
- › Clinical trials recording demographic **and** genetic markers
- › Any design that takes interactions seriously —  $p$  grows fast

**The problem:** Lin's fix fits OLS **separately in each arm**, inverting an arm-level  $X'X$ . Once  $p$  approaches  $N_1$  or  $N_0$ , that regression **“can overfit the data badly, or may even be ill posed”**.



# The Same Estimator, a Different Fit

Lin's arm-wise estimator:

$$\hat{\tau}_{\text{Lin}} = \left[ \bar{Y}_1 - (\bar{X}_1 - \bar{X})' \hat{\beta}^{(1)} \right] - \left[ \bar{Y}_0 - (\bar{X}_0 - \bar{X})' \hat{\beta}^{(0)} \right]$$

**The proposal:** fit each arm's  $\hat{\beta}^{(a)}$  by the Lasso, for  $a \in \{0, 1\}$ :

$$\hat{\beta}_{\text{Lasso}}^{(a)} = \arg \min_{\beta} \frac{1}{2N_a} \sum_{i:A_i=a} \left( Y_i - \bar{Y}_a - (X_i - \bar{X}_a)' \beta \right)^2 + \lambda_a \|\beta\|_1$$

**What changed:** nothing but the fit.

- › Same estimand, same design, same centering, same arm-wise structure
- › **Only how  $\hat{\beta}^{(a)}$  is estimated** — choose  $\lambda_a$  by cross-validation



# The Penalty Brings Freedman Back

Bloniarz et al. are explicit about when adjustment is free:

if the adjustment vectors [are] fixed (nonrandom), or if they are derived from an **independent data source**, then this estimator is still unbiased

This is **Aronow & Middleton's condition**,  $\text{Cov}(A_i, f_i) = 0$ , **restated**. But the Lasso fits  $\hat{\beta}^{(a)}$  on the same outcomes we are about to average:

- › Weights are no longer outcome-independent  $\Rightarrow$  exact unbiasedness is gone
- › Lin's OLS version has bias vanishing at rate  $1/n$
- › The Lasso needs **sparsity** before it can say anything at all

**Key Insight:** Freedman's objection has **moved**, not been answered — the risk now lives in the fitting procedure, not the functional form.



# Proof Sketch: Why the Lasso Adjustment Still Works

**Setup:** Compare  $\hat{\tau}_{\text{Lasso}}$  to the **oracle**  $\hat{\tau}_{\text{orc}}$ , which plugs in fixed  $\beta^{(1)}, \beta^{(0)}$ .

**Key Steps:**

1. **The oracle is unbiased:** its adjustment never looks at  $Y$ , so Aronow & Middleton applies directly
2. **Define:**  $\Delta^{(a)} \equiv \hat{\beta}^{(a)} - \beta^{(a)}$ ,  $d_a \equiv \bar{X}_a - \bar{X}$ , and  $s \equiv$  the number of nonzero  $\beta^{(a)}$
3. **Isolate the cost:**  $\hat{\tau}_{\text{Lasso}} - \hat{\tau}_{\text{orc}} = -d_1' \Delta^{(1)} + d_0' \Delta^{(0)}$
4. **Split with Hölder:**  $|d_a' \Delta^{(a)}| \leq \|d_a\|_{\infty} \|\Delta^{(a)}\|_1$
5. **Bound each factor:**  $\|d_a\|_{\infty} = O_P(\sqrt{(\log p)/n})$  by randomization,  $\|\Delta^{(a)}\|_1 = O_P(s \sqrt{(\log p)/n})$  by sparsity
6. **Multiply:** the product is  $(s \log p)/n$ , against a standard error of order  $1/\sqrt{n}$  — negligible when  $s \log p \ll \sqrt{n}$

**Key Insight:** Neither factor has to be small on its own — only their **product** has to beat the noise.



# What You Actually Get

**Theorem 1:** the Lasso-adjusted estimator is asymptotically normal:

$$\sqrt{n} \left( \hat{\tau}_{\text{Lasso}} - \tau \right) \rightarrow \mathcal{N}(0, \sigma^2)$$

**Corollary 1:** its asymptotic variance is **no worse than unadjusted**.

**Inference:** intervals are **Neyman-type conservative**, exactly as in Week 1.

- ›  $\sigma_{e^{(1)}e^{(0)}}$  cannot be consistently estimated
- › We never see both potential outcomes for the same unit



# What It Takes

## Three conditions:

- › **Sparsity:** only  $s$  of the  $p$  covariates matter, with  $s \log p \ll \sqrt{n}$
- › **Moment conditions** on the potential outcomes
- › **Enough independent variation:** no small group of covariates is nearly a combination of the others, or the Lasso cannot tell which ones to select

## Their own caveats:

- › “Both assumptions are difficult to check in practice”
- › In their own application  $(s \log p) / \sqrt{n} = 3.55$  — “**not close to zero**”
- › They recommend `cv(Lasso+OLS)` in practice: Lasso to select, OLS to adjust



(Bai, Shaikh, and Tabord-Meehan 2024):  
Taking a step back

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# Fixed Units, or Draws?

**Design-based** — everything today:  $y_i(1)$ ,  $y_i(0)$  are fixed, and the only randomness is  $A$ .

**Super-population** — Bai, Shaikh & Tabord-Meehan: units are **i.i.d. draws**, so the potential outcomes are random too.

- › It approximates a finite population from which a *negligible fraction* is sampled
- › But results derived that way are useful “even outside of this limiting case”
- › The framing decides **what is random**, and so what a standard error is estimating

**Their example:** Lin’s regression centres at  $\bar{X}$  — a fixed feature of the sample design-based, but **random** once the sample is drawn from a population, so its robust SE is not automatically valid.

**The rest of the survey:** stratification (blocks, matched pairs), adjustment when the assignment itself is stratified, cluster randomisation, and short takes on heterogeneous effects, interference, multiple testing, policy learning. Companion R package **sreg**.



# Synthesis

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# What To Do

1. **Adjust by default, Lin-style** — centre the covariates, include treatment-by-covariate interactions, use HC2 standard errors.
2. **Pre-specify the adjustment** before seeing outcomes, and **report the unadjusted estimate** alongside it for comparison.
3. **When  $p$  is large**, fit the adjustment arm-wise with the Lasso (or `cv(Lasso+OLS)` ) and accept conservative intervals.
4. **Keep the estimand fixed.** Every method today estimates the same ATE under the same randomisation — they differ only in how the adjustment is fit.

Adjustment buys precision, and pays for it in model dependence. The design tells you the price is *small* — it never tells you it is *zero*.



# More to read

| Paper                                                      | Why you'd read it                                                                                                           |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| ( <a href="#">Wager et al. 2016</a> )                      | Cross-estimation — fit the adjustment out-of-fold and finite-sample unbiasedness comes <i>back</i> , for any ML adjustment. |
| ( <a href="#">Belloni, Chernozhukov, and Hansen 2014</a> ) | Post-double-selection: the origin of OLS after Lasso.                                                                       |
| ( <a href="#">Tsiatis et al. 2008</a> )                    | The semiparametric/ AnCova framing of the same problem, from the clinical-trials literature.                                |
| ( <a href="#">Hill 2011</a> )                              | Skip the linear adjustment entirely: fit a flexible Bayesian model, impute both potential outcomes, average.                |
| ( <a href="#">Bannick et al. 2025</a> )                    | Adjustment when the <i>randomisation itself</i> is covariate-adaptive.                                                      |

# References

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